

Lahore College for Women University, Lahore
Final Term Examination

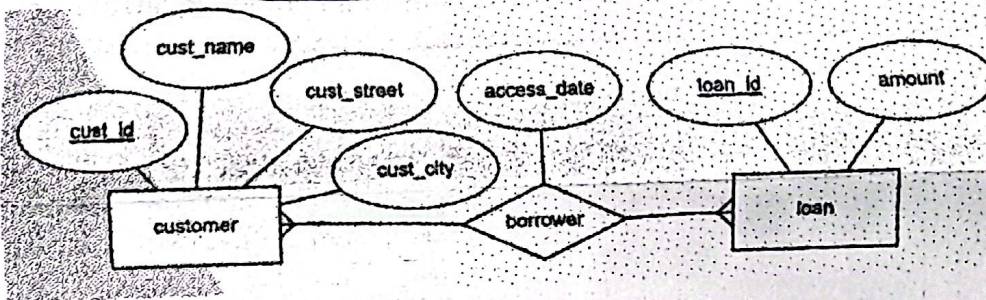
Course Title: Database Systems
Maximum marks: 50 (Time: 3 Hrs)
Q1)

Program: BSCS (IV)
Session (2023-2027)
(5+5)

a) Convert the following table into 3NF.

OrderProduct Table												
Order No	Prod No	Cust No	Name	Addr	City	St	Zip	Order Date	Promised Date	Desc	Qty Ord	Unit Price
61384	A128	1273	Cont. Designs	123 Oak	Austin	TX	78746	11/04/15	11/21/15	Bookcase	4	200
61384	B381	1273	Cont. Designs	123 Oak	Austin	TX	78746	11/04/15	11/21/15	Cabinet	2	150
61384	R210	1273	Cont. Designs	123 Oak	Austin	TX	78746	11/04/15	11/21/15	Table	1	500
62890	W891	3891	J Consultants	523 Pine	Waco	TX	76712	11/15/15	11/21/15	Chair	2	300
62890	A128	3891	J Consultants	523 Pine	Waco	TX	76712	11/15/15	11/21/15	Bookcase	8	200
63129	W891	1273	Cont. Designs	123 Oak	Austin	TX	78746	12/10/15	12/29/15	Chair	6	300
64000	B381	4545	ABC Ins Co	555 Elm	OKC	OK	73134	04/10/15	05/17/15	Cabinet	1	150
64000	A128	4545	ABC Ins Co	555 Elm	OKC	OK	73134	04/10/15	05/17/15	Bookcase	1	200

b) Convert the following ER diagram into Relation.



Q2) Define the following with one example of each concept. (10)

- Derived Attribute
- Unary Relationship
- Associative Entity — *used to show Many-to-many relation*
- Completeness Constraints —
- Weak Entity

Q3) Write down the following SQL queries. Consider the following Schema. (8*2)

Customers(customer_id, name, city)
Orders(order_id, customer_id, order_date, total_amount)
Products(product_id, name, price)
OrderItems(order_id, product_id, quantity)

- List all customers from 'New York'.
- Show each order with the customer name.
- Find total amount spent by each customer.

Agenda
14 Nov 2023

P-T-O

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4. List all products with their total quantity sold.
5. Find the most expensive product ordered.
6. Get customers who placed more than 3 orders.
7. Show orders with no items.
8. List top 3 customers by total spending.

Q4)

(4)

Considering the hierarchy of your university, employee can be of two types. Teaching & non-teaching. Draw separate EER segment for each of the following situations

- a) At a given time, an employee must be exactly one of these subtypes.
- b) An employee may or may not be one of these subtypes but at a given time, employee cannot be part of more than one subtype.
- c) An employee must be a member of one of these subtypes but at the same time an employee can perform any non-teaching task assigned by the management.

Ayesha

14/11/2023